Math 5601: Introduction to Numerical Analysis
Homework assignment 7

Show all relevant work in detail to justify your conclusions. Partial credit depends upon the work you show.

Problem #1: Find the Gaussian quadrature of three nodes for $\int_{-1}^{1} f(x)\,dx$ by requiring it to be exact for $f(x) = 1, x, x^2, x^3, x^4, x^5$. If needed, you may use the matlab command "solve" to obtain the solution of a system of nonlinear equations.

Problem #2: Find the Gaussian quadrature of three nodes for $\int_{-1}^{1} f(x)\,dx$ by using orthogonal polynomials.

Problem #3: Suppose that $I(h)$ is an approximation to $\int_a^b f(x)dx$ where h is the width of the uniform subdivision of $[a, b]$. Suppose that the error satisfies

$$I(h) - \int_a^b f(x)dx = c_1 h + c_2 h^2 + O(h^3)$$

where c_1 and c_2 are constants independent of h. Suppose $I(h)$, $I(\frac{h}{2})$, $I(\frac{h}{3})$ have been calculated. Find an $O(h^3)$ approximation to $\int_a^b f(x)dx$.

Math 5601 - Homework 07 - Tyler Glidden

Problem 1

Find Gaussian Quadrature for 3 nodes $\int_{-1}^{1} F(x)dx$. Require it to be exact for

$F(x) = 1, x, x^2, x^3, x^4, x^5$.

$m=2, \quad 2m+1 = 5$

$$\int_{-1}^{1} F(x)\,dx \approx \sum a_j F(x_j)$$

$$a_0 + a_1 + a_2 = \int_{-1}^{1} 1\,dx = 2$$

$$\begin{cases} a_0x_0 + a_1x_1 + a_2x_2 = \int_{-1}^{1} x\,dx = 0 \\ a_0x_0^2 + a_1x_1^2 + a_2x_2^2 = \int_{-1}^{1} x^2\,dx = 2/3 \\ a_0x_0^3 + a_1x_1^3 + a_2x_2^3 = \int_{-1}^{1} x^3\,dx = 0 \\ a_0x_0^4 + a_1x_1^4 + a_2x_2^4 = \int_{-1}^{1} x^4\,dx = 2/5 \\ a_0x_0^5 + a_1x_1^5 + a_2x_2^5 = \int_{-1}^{1} x^5\,dx = 0 \end{cases}$$

Solve in MatLab using

Fsolve

$a_0 = 0.5556$ $\quad x_0 = -0.7746$

$a_1 = 0.8889$ $\quad x_1 = 0$

$a_2 = 0.5556$ $\quad x_2 = 0.7746$

$$\Rightarrow \int_{-1}^{1} F(x)\,dx \approx 0.5556\,F(-0.7746) + 0.8889\,f(0) + 0.5556\,f(0.7746)$$

Remark:

After completing Problem 2, these decimals match exactly to

$$5/9\,F\left(-\sqrt{3/5}\right) + \frac{8}{9}\,F(0) + 5/9\,F\left(\sqrt{3/5}\right).$$

Problem 2

Find Gaussian Quadrature for 3 nodes by using Orthoganal Polynomials

Use Gram-Schmidt

$m=2 \Rightarrow$ we need $P_0(x), P_1(x), P_2(x), P_3(x)$

$u_1 = 1 \quad u_2 = x \quad u_3 = x^2 \quad u_4 = x^3$

$V_1 = u_1 = 1 //$

$V_2 = u_2 - \frac{(u_2, V_1)}{(V_1, V_1)} \cdot V_1 = x //$ ($(u_2,V_1)=0$)

$V_3 = u_3 - \frac{(u_3,V_2)}{(V_2,V_2)} V_2 - \frac{(u_3,V_1)}{(V_1,V_1)} V_1$ ($(u_3,V_2)=0$)

$= x^2 - \frac{\frac{2}{3}}{2} \cdot 1 = x^2 - \frac{1}{3} //$

$V_4 = u_4 - \frac{(u_4,V_3)}{(V_3,V_3)} V_3 - \frac{(u_4,V_2)}{(V_2,V_2)} V_2 - \frac{(u_4,V_1)}{(V_1,V_1)} V_1$ ($(u_4,V_3)=0$, $(u_4,V_1)=0$)

$= x^3 - \frac{\frac{2}{5}}{\frac{2}{3}} \cdot x = x^3 - \frac{3}{5}x //$

Scratch Area

$(u_2,V_1) = \int_{-1}^{1} x \cdot 1 \, dx = 0$

$(u_3,V_2) = \int_{-1}^{1} x^2 \cdot x \, dx = \int_{-1}^{1} x^3 \, dx = 0$

$(u_3,V_1) = \int_{-1}^{1} x^2 \cdot 1 \, dx = 2/3 //$

$(V_1,V_1) = \int_{-1}^{1} 1 \cdot 1 \, dx = 2 //$

$(u_4,V_3) = \int_{-1}^{1} x^3 (x^2 - 1/3) \, dx = \int_{-1}^{1} (x^5 - x^3/3) \, dx$ → Both odd functions, $= 0$

$(u_4,V_2) = \int_{-1}^{1} x^3 \cdot x \, dx = \int_{-1}^{1} x^4 = 2/5$

$(V_2,V_2) = \int_{-1}^{1} x \cdot x \, dx = \int_{-1}^{1} x^2 \, dx = 2/3$

$(u_4,V_1) = \int_{-1}^{1} x^3 \cdot 1 \, dx = 0$

$x^2 - 1 = (x+1)(x-1)$

nodes:

$x^3 - 3/5 x \overset{\text{set}}{=} 0$

$x = 0$

$x = \pm\sqrt{3/5}$

$x_0 = -\sqrt{3/5}$

$x_1 = 0$

$x_2 = +\sqrt{3/5}$

Weights:

$a_j = \int_a^b L_j(x) \rho(x) \, dx$

$a_0 = \int_{-1}^{1} L_0(x) \, dx = \int_{-1}^{1} \frac{x - x_1}{x_0 - x_1} \cdot \frac{x - x_2}{x_0 - x_2} \, dx$

$\int_{-1}^{1} \frac{x - 0}{-\sqrt{3/5} - 0} \cdot \frac{x - \sqrt{3/5}}{-\sqrt{3/5} - \sqrt{3/5}} \, dx = \int_{-1}^{1} \frac{x}{-\sqrt{3/5}} \cdot \frac{x - \sqrt{3/5}}{-2\sqrt{3/5}} \, dx = \int_{-1}^{1} \frac{x^2 - x\sqrt{3/5}}{2 \cdot \frac{3}{5}} \, dx$ ($x\sqrt{3/5}$ → odd function, $= 0$)

$= \int_{-1}^{1} \frac{5x^2}{6} \, dx = \frac{5}{6} \cdot \frac{2}{3} = \frac{10}{18} // = 5/9 //$

Diff of ^2

$a_1 = \int_{-1}^{1} L_1(x) \, dx = \int_{-1}^{1} \frac{x - x_0}{x_1 - x_0} \cdot \frac{x - x_2}{x_1 - x_2} \, dx = \int_{-1}^{1} \frac{(x + \sqrt{3/5})}{0 + \sqrt{3/5}} \cdot \frac{(x - \sqrt{3/5})}{0 - \sqrt{3/5}} \, dx = \int_{-1}^{1} \frac{x^2 - 3/5}{-\frac{3}{5}} \, dx$

$= -\frac{5}{3} \cdot \int_{-1}^{1} x^2 - 3/5 \, dx = -\frac{5}{3}\left(\frac{x^3}{3} - \frac{3}{5}x\right)\Big|_{-1}^{1} = -\frac{5}{3}\left(\frac{1}{3} - \frac{3}{5} - \frac{-1}{3} + \frac{+3}{5}\right) = -\frac{5}{3}\left(\frac{2}{3} - \frac{6}{5}\right) = -\frac{5}{3}\left(\frac{10}{15} - \frac{18}{15}\right)$

$= -\frac{5}{3}\left(-\frac{8}{15}\right) = \frac{40}{45} = 8/9 //$

$a_2 = \int_{-1}^{1} L_2(x) \, dx = \int_{-1}^{1} \frac{x - x_0}{x_2 - x_0} \cdot \frac{x - x_1}{x_2 - x_1} \, dx = \int_{-1}^{1} \frac{x + \sqrt{3/5}}{\sqrt{3/5} + \sqrt{3/5}} \cdot \frac{x - 0}{\sqrt{3/5} - 0} \, dx = \int_{-1}^{1} \frac{(x + \sqrt{3/5})\, x}{2\sqrt{3/5}} \, dx$

$= \int_{-1}^{1} \frac{x^2 + \sqrt{3/5}\, x}{2 \cdot 3/5} \, dx$ ($\sqrt{3/5}\, x$ → odd, $= 0$) $= 5/9 //$ (similar to a_0)

$\Rightarrow Q(f) = 5/9 \, f\left(-\sqrt{3/5}\right) + \frac{8}{9} f(0) + 5/9 \, f\left(+\sqrt{3/5}\right) \approx \int_{-1}^{1} f(x) \, dx.$

Problem 3

Suppose $I(h)$ is an approximation:

① $I(h) - \int_a^b f(x)dx = C_1 h + C_2 h^2 + O(h^3)$

② $I(h/2) - \int_a^b f(x)dx = C_1 h/2 + C_2 h^2/4 + O(h^3)$

③ $I(h/3) - \int_a^b f(x)dx = C_1 h/3 + C_2 h^2/9 + O(h^3)$

$a_1 ① + a_2 ② + a_3 ③$

$\Leftrightarrow a_1 I(h) + a_2 I(h/2) + a_3 I(h/3) = (a_1 + a_2 + a_3)\int_a^b f(x)dx$

$+ C_1 h(a_1 + \frac{1}{2}a_2 + \frac{1}{3}a_3)$

$+ C_2 h^2(a_1 + \frac{1}{4}a_2 + \frac{1}{9}a_3)$

$+ O(h^3)$

$$\begin{cases} a_1 + a_2 + a_3 \overset{\text{set}}{=} 1 \\ a_1 + \frac{1}{2}a_2 + \frac{1}{3}a_3 \overset{\text{set}}{=} 0 \\ a_1 + \frac{1}{4}a_2 + \frac{1}{9}a_3 \overset{\text{set}}{=} 0 \end{cases}$$

$a_1 = 1/2$

$a_2 = -4$

$a_3 = 4.5$

$\Rightarrow \frac{1}{2} I(h) - 4 I(h/2) + 4.5 I(h/3) = \int_a^b F(x)dx + O(h^3)$